5. Other measures and dimensions

The main part of this chapter will deal with Minkowski and packing dimensions and packing measures and their relations to Hausdorff measures. We begin with two slight modifications of Hausdorff measures.

Spherical measures

5.1. Let $0 \le t < \infty$. If we apply Carathéodory's construction 4.1 taking $\mathcal{F}$ to be the family of all closed (or open) balls in a separable metric space X and $\zeta(B) = d(B)^t$, the resulting measure $\psi(\mathcal{F}, \zeta)$ is called t-dimensional *spherical measure*. We denote it by $\mathcal{S}^t$. In $\mathbf{R}^n$, for $n \ge 2$ and $0 < t < n$, it differs from the t-dimensional Hausdorff measure, but they are related by the inequalities

$$\mathcal{H}^t(A) \le \mathcal{S}^t(A) \le 2^t \mathcal{H}^t(A).$$

The left hand inequality follows immediately from the definitions and the right one from the fact that any bounded set $E \subset X$ is contained in a ball of diameter $2d(E)$. Hence for example for finding the Hausdorff dimension of a given set, we can use spherical measures and coverings with balls in place of Hausdorff measures and more general coverings.

We give an example of a compact set S in $\mathbf{R}^2$ for which $\mathcal{H}^t(S) < \mathcal{S}^t(S)$. The self-similar set indicated by Figure 5.1, a Sierpinski gasket, suffices. Besicovitch [1, § 47] studied in detail for $t = 1$ a modified example giving the biggest possible ratio $\mathcal{S}^1(A)/\mathcal{H}^1(A) = 2/\sqrt{3}$.

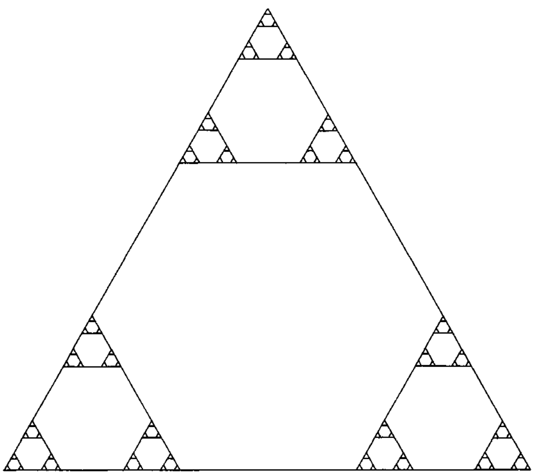

Figure 5.1. A Sierpinski gasket.

Net measures

5.2. The *net measures* are denoted by $\mathcal{N}^t$ and they are obtained from the Carathéodory construction in $\mathbf{R}^n$ by taking again $\zeta(E) = d(E)^t$ and as $\mathcal{F}$ the family of half-open dyadic cubes in $\mathbf{R}^n$, that is, cubes of the form

$$\{x \in \mathbf{R}^n : k_i\, 2^{-m} \leq x_i < (k_i + 1)\, 2^{-m} \quad \text{for } i = 1, \ldots, n\}$$

where k_i and m are arbitrary integers. The net measures are often easier to handle than Hausdorff measures, because every family $\mathcal{A}$ of such dyadic cubes with $\sup_{Q \in \mathcal{A}} d(Q) < \infty$ has a disjoint subfamily with the same union; select those cubes in $\mathcal{A}$ which are not contained in any other. As for spherical measures one gets

$$\mathcal{H}^t(A) \leq \mathcal{N}^t(A) \leq 4^t n^{t/2}\, \mathcal{H}^t(A).$$

For applications of net measures to Hausdorff measures, see e.g. Falconer [4, Chapter 5] and Rogers [1, Chapter 2].

The Hausdorff dimension is a natural parameter to measure the metric size of any given set in a metric space. However, it is not the only one. There are other parameters whose use is well justified from the point of view both of the geometric contents of the very definitions and of the applications.

Minkowski dimensions

5.3. The Hausdorff dimension is defined by looking at the coverings of a set by small sets E_i and inspecting the sums $\sum d(E_i)^s$. As noted before the sets E_i could be arbitrary or they could be balls or, in $\mathbf{R}^n$, dyadic cubes. One of the most immediate modifications from this leads to coverings with balls, for example, of the same size. Although the following makes sense in any metric space, we restrict attention to $\mathbf{R}^n$.

Let A be a non-empty bounded subset of $\mathbf{R}^n$. For $0 < \varepsilon < \infty$, let $N(A, \varepsilon)$ be the smallest number of ε-balls needed to cover A:

$$N(A, \varepsilon) = \min \left\{ k : A \subset \bigcup_{i=1}^{k} B(x_i, \varepsilon) \quad \text{for some } x_i \in \mathbf{R}^n \right\}.$$

The *upper* and *lower Minkowski dimensions* of A are defined by

$$\overline{\dim}_M A = \inf\{s : \limsup_{\varepsilon \downarrow 0} N(A, \varepsilon)\, \varepsilon^s = 0\}$$

and

$$\underline{\dim}_M A = \inf\{s : \liminf_{\varepsilon \downarrow 0} N(A,\varepsilon)\,\varepsilon^s = 0\}.$$

It is obvious that

$$\begin{aligned}\overline{\dim}_M A &= \inf\{s : \limsup_{\varepsilon \downarrow 0} N(A,\varepsilon)\,\varepsilon^s < \infty\}\\ &= \sup\{s : \limsup_{\varepsilon \downarrow 0} N(A,\varepsilon)\,\varepsilon^s = \infty\}\\ &= \sup\{s : \limsup_{\varepsilon \downarrow 0} N(A,\varepsilon)\,\varepsilon^s > 0\},\end{aligned}$$

and similarly for $\underline{\dim}_M A$.

It follows immediately from the definitions that

$$\dim A \le \underline{\dim}_M A \le \overline{\dim}_M A \le n,$$

and these inequalities can be strict. For the left inequality one can get an example even from countable compact sets. For instance,

$$\underline{\dim}_M\big(\{0\} \cup \{1/i : i = 1, 2, \dots\}\big) = 1/2.$$

We leave the proof as an exercise.

We now briefly indicate how to construct a compact set $E \subset \mathbf{R}^1$ with $\underline{\dim}_M E < \overline{\dim}_M E$. Let $0 < s < t < 1$. As in 4.10, start constructing a Cantor set $C(\lambda)$ of Hausdorff dimension less than s, i.e. $s > \log 2/\log(1/\lambda)$. Thus we have two subintervals $I_{1,1}$ and $I_{1,2}$ whose lengths d_1 satisfy $2\,d_1^s \le 1$. In each $I_{1,j}$ perform now the construction of $C(\mu)$ of dimension greater than t sufficiently many times so that you will have altogether 2^{k_2} subintervals $I_{2,1}, \dots, I_{2,2^{k_2}}$ of $[0,1]$ whose lengths d_2 satisfy $2^{k_2} d_2^t \ge 1$. After that continue again with the construction of $C(\lambda)$ and so on. The resulting Cantor set of this process will have the lower Minkowski dimension at most s and the upper at least t. We omit the details.

To obtain a compact set $E \subset \mathbf{R}^1$ with $0 < s = \dim E < \underline{\dim}_M E = t < 1$, perform a Cantor construction where inside the intervals I already selected one chooses many intervals I_j of different sizes such that $\sum_j d(I_j)^s = d(I)^s$ but $N(\bigcup_j I_j, \varepsilon)\,\varepsilon^t \ge 1$ for all $0 < \varepsilon \le d(I)$. A combination and modification of these ideas shows that for any $0 \le s \le t \le u \le 1$ there is a compact set $E \subset \mathbf{R}^1$ with $\dim E = s$, $\underline{\dim}_M E = t$ and $\overline{\dim}_M E = u$.

There are some obvious equivalent definitions of Minkowski dimensions. For example,

$$\overline{\dim}_M A = \limsup_{\varepsilon \downarrow 0} \frac{\log N(A,\varepsilon)}{\log(1/\varepsilon)},$$
$$\underline{\dim}_M A = \liminf_{\varepsilon \downarrow 0} \frac{\log N(A,\varepsilon)}{\log(1/\varepsilon)}.$$

The proofs are left as exercises. The corresponding formulas can be given also in terms of the packing numbers $P(A,\varepsilon)$ instead of the covering numbers $N(A,\varepsilon)$. Let $P(A,\varepsilon)$ be the greatest number of disjoint ε-balls with centres in A:

$$P(A,\varepsilon) = \max\{k : \text{there are disjoint balls } B(x_i,\varepsilon), \\ i=1,\dots,k, \text{ with } x_i \in A\}.$$

Then

$$N(A,2\varepsilon) \le P(A,\varepsilon) \le N(A,\varepsilon/2). \tag{5.4}$$

To verify the first inequality, let $k = P(A,\varepsilon)$ and choose disjoint balls $B(x_i,\varepsilon)$, $x_i \in A$, $i=1,\dots,k$. If there exists $x \in A \setminus \bigcup_{i=1}^k B(x_i,2\varepsilon)$, the balls $B(x_1,\varepsilon),\dots,B(x_k,\varepsilon),B(x,\varepsilon)$ would be disjoint giving $k+1 \le P(A,\varepsilon) = k$. Hence the balls $B(x_i,2\varepsilon)$ cover A, and so $N(A,2\varepsilon) \le k = P(A,\varepsilon)$.

For the second inequality let $N = N(A,\varepsilon/2)$ and $k = P(A,\varepsilon)$, and choose $x_1,\dots,x_N \in \mathbf{R}^n$, $y_1,\dots,y_k \in A$ such that $A \subset \bigcup_{i=1}^N B(x_i,\varepsilon/2)$ and the balls $B(y_j,\varepsilon)$, $j=1,\dots,k$, are disjoint. Then each y_j belongs to some $B(x_i,\varepsilon/2)$ and no $B(x_i,\varepsilon/2)$ contains more than one point y_j, the balls $B(y_j,\varepsilon)$ being disjoint. Thus $k \le N$, which gives $P(A,\varepsilon) \le N(A,\varepsilon/2)$.

The inequalities (5.4) give immediately the formulas for the Minkowski dimensions in terms of $P(A,\varepsilon)$. For example,

$$\overline{\dim}_M A = \limsup_{\varepsilon \downarrow 0} \frac{\log P(A,\varepsilon)}{\log(1/\varepsilon)}.$$

The Minkowski dimensions can also easily be seen to be determined with dyadic cubes: let $\widetilde{N}_m(A)$ be the number of dyadic cubes ("boxes") of side-length 2^{-m} which meet A. Then

$$\overline{\dim}_M A = \limsup_{m \to \infty} \frac{\log \widetilde{N}_m(A)}{m \log 2}.$$

This formulation has led to the term box counting dimension for $\overline{\dim}_M$ (or $\underline{\dim}_M$), nowadays widely used especially by experimentalists, who want to compute the dimension by counting the boxes, see Falconer [16] and Feder [1]. Minkowski dimensions have also many other names. They are sometimes called metric, fractal or capacity dimensions. From our point of view the last term is very misleading, since the potential-theoretic capacity and capacitary dimension, which we shall introduce in Chapter 8, are quite different things.

The term Minkowski dimension really has a rather different origin, namely in terms of the Minkowski contents.

5.5. Minkowski contents. Let A be a non-empty bounded subset of $\mathbf{R}^n$. Recall that for $0 < \varepsilon < \infty$ the closed ε-neighbourhood of A is

$$A(\varepsilon) = \{x \in \mathbf{R}^n : d(x, A) \leq \varepsilon\}.$$

The Lebesgue measure of $A(\varepsilon)$ can almost be given in terms of the covering and packing numbers:

$$P(A,\varepsilon)\,\alpha(n)\,\varepsilon^n \leq \mathcal{L}^n(A(\varepsilon)) \leq N(A,\varepsilon)\,\alpha(n)(2\varepsilon)^n. \tag{5.6}$$

These inequalities follow immediately from the facts that any union of ε-balls with centres in A is contained in $A(\varepsilon)$, and any union of (2ε)-balls covers $A(\varepsilon)$ if the union of the corresponding ε-balls covers A. Hence it is clear that the Minkowski dimensions of A depend on the behaviour of $\mathcal{L}^n(A(\varepsilon))$ as $\varepsilon \downarrow 0$. To formulate this, define the s-dimensional upper and lower Minkowski contents of A by

$$\mathcal{M}^{*s}(A) = \limsup_{\varepsilon\downarrow 0}(2\varepsilon)^{s-n}\mathcal{L}^n(A(\varepsilon)),$$
$$\mathcal{M}^{s}_{*}(A) = \liminf_{\varepsilon\downarrow 0}(2\varepsilon)^{s-n}\mathcal{L}^n(A(\varepsilon)).$$

Then

$$\overline{\dim}_M A = \inf\{s : \mathcal{M}^{*s}(A) = 0\} = \sup\{s : \mathcal{M}^{*s}(A) > 0\}$$

and

$$\underline{\dim}_M A = \inf\{s : \mathcal{M}^{s}_{*}(A) = 0\} = \sup\{s : \mathcal{M}^{s}_{*}(A) > 0\}.$$

The Minkowski contents are not measures as they are not subadditive. By (5.4) and (5.6) we have $2^{-s-n}\alpha(n)\,\mathcal{H}^s(A) \leq \mathcal{M}^s_*(A)$ which, except for improving the constant, is about all we can say on their general relations to Hausdorff measures. But for nice sets both Minkowski contents equal a constant multiple of the Hausdorff measure. For example if Γ is a rectifiable curve, then $\mathcal{M}^{*1}(\Gamma) = \mathcal{M}^1_*(\Gamma) = \mathcal{H}^1(\Gamma)$. For more general results, see Federer [3, 3.2.37–44].

The behaviour of the covering and packing numbers $N(A,\varepsilon)$ and $P(A,\varepsilon)$ as $\varepsilon \downarrow 0$ was studied in Lalley [1] for self-similar sets and in more general situations applying also to discrete groups, recall 4.16, in Lalley [2]. Lapidus and Pomerance [1] characterized the compact subsets of $\mathbf{R}$ for which the upper and lower Minkowski contents agree and Lapidus and Maier [1] derived a connection of this to the Riemann hypothesis, see also Falconer [22]. This is related to the question of the distribution of the eigenvalues of the Laplacian on domains with fractal boundaries, which is studied also by Evans and Harris [1]. For some other geometric questions related to Minkowski content and dimension, see Martio and Vuorinen [1], Mattila and Vuorinen [1], Salli [2] and Tricot [4]–[6]. Harrison and Norton [1]–[2] developed a theory for integrating Hölder continuous differential forms over fractal boundaries. They proved a general Gauss–Green theorem in which the Minkowski dimension of the boundary and the Hölder continuity exponent of the admissible forms are delicately related.

Recently there has been a lot of interest in the question, when do the Hausdorff and Minkowski dimensions agree. Often this is a consequence of the existence of a sufficiently regular measure. The following simple proof was given by Salli [2]. A more general formulation was found by Young [1].

5.7. Theorem. *Let A be a non-empty bounded subset of $\mathbf{R}^n$. Suppose there are a Borel measure μ on $\mathbf{R}^n$ and positive numbers a, b, r_0 and s such that $0 < \mu(A) \le \mu(\mathbf{R}^n) < \infty$ and*

$$0 < ar^s \le \mu(B(x,r)) \le br^s < \infty \quad \text{for } x \in A,\ 0 < r \le r_0.$$

Then $\dim A = \underline{\dim}_M A = \overline{\dim}_M A = s$.

Proof. If A is covered by sets E_i such that $0 < d(E_i) \le r_0$ and $A \cap E_i \ne \emptyset$, we can pick points $x_i \in A \cap E_i$ and A is then also covered by the balls $B(x_i, d(E_i))$. Thus

$$b \sum_i d(E_i)^s \ge \sum_i \mu\big(B(x_i, d(E_i))\big) \ge \mu(A) > 0.$$

This gives $\mathcal{H}^s(A) \ge \mu(A)/b$, whence

$$s \le \dim A \le \underline{\dim}_M A \le \overline{\dim}_M A.$$

On the other hand, let $0 < \varepsilon \leq r_0$, $k = P(A,\varepsilon)$, and choose disjoint balls $B(x_i,\varepsilon)$, $x_i \in A$, $i = 1,\ldots,k$. Then

$$aP(A,\varepsilon)\,\varepsilon^s \leq \sum_{i=1}^{k} \mu(B(x_i,\varepsilon)) \leq \mu(\mathbf{R}^n),$$

which implies $\overline{\dim}_M A \leq s$. □

Combining this with Hutchinson's result Theorem 4.14, we obtain

5.8. Corollary. *Let K be a self-similar set generated by similitudes for which the open set condition holds. Then $\overline{\dim}_M K = \dim K$.*

In fact, according to a result of Falconer [15] $\overline{\dim}_M K = \dim K$ for any compact set K which is invariant under a finite family of similitudes or even contractive conformal transformations. In particular, no separation condition is needed.

Many of the references in 4.15 also deal with the Minkowski dimension and compare it to the Hausdorff dimension for self-affine sets and attractors of dynamical systems. Often these dimensions differ for specific self-affine sets, see Gatzouras and Lalley [1], Kenyon and Peres [2] and Lalley and Gatzouras [1], but in some cases they agree "generically", see Falconer [12], and also Bedford and Urbański [1].

Packing dimensions and measures

5.9. Packing dimensions. Earlier we observed that even a compact countable set can have positive Minkowski dimension. This is a reflection of the fact that the Minkowski dimensions are lacking one of the fundamental properties of the Hausdorff dimension:

$$\dim\left(\bigcup_{i=1}^{\infty} A_i\right) = \sup\{\dim A_i : i = 1,2,\ldots\}.$$

We can easily modify the Minkowski dimensions to arrive at dimensions which have this property. We call them *upper* and *lower packing dimensions* and they can be defined for any subset A of $\mathbf{R}^n$ by

$$\overline{\dim}_p A = \inf\left\{\sup_i \overline{\dim}_M A_i : A = \bigcup_{i=1}^{\infty} A_i,\ A_i \text{ is bounded}\right\},$$

$$\underline{\dim}_p A = \inf\left\{\sup_i \underline{\dim}_M A_i : A = \bigcup_{i=1}^{\infty} A_i,\ A_i \text{ is bounded}\right\}.$$

Clearly,

$$\dim A \le \underline{\dim}_p A \le \underline{\dim}_M A$$

and

$$\underline{\dim}_p A \le \overline{\dim}_p A \le \overline{\dim}_M A.$$

All these inequalities can be strict, see Tricot [2] for some examples. But now $\overline{\dim}_p A = 0$ for all countable sets. In 6.13 we shall give a condition weaker than that of Theorem 5.7 which guarantees that $\overline{\dim}_p A$ will equal $\dim A$.

The upper packing dimension can also be defined in terms of the packing measures, which we now introduce. Because of this the upper packing dimension is often called just packing dimension.

5.10. Packing measures. Let $0 \le s < \infty$. For $A \subset \mathbf{R}^n$ and $0 < \delta < \infty$, put

$$P^s_\delta(A) = \sup \sum_i d(B_i)^s$$

where the supremum is taken over all disjoint families (packings) of closed balls $\{B_1, B_2, \dots\}$ such that $d(B_i) \le \delta$ and the centres of the B_i's are in A. Then $P^s_\delta(A)$ is non-decreasing with respect to δ and we set

$$P^s(A) = \lim_{\delta \downarrow 0} P^s_\delta(A) = \inf_{\delta > 0} P^s_\delta(A).$$

Obviously P^s is monotonic and $P^s(\emptyset) = 0$, but unfortunately it is not countably subadditive. To get a measure out of it we use a standard procedure and define

$$\mathcal{P}^s(A) = \inf\Big\{\sum_{i=1}^\infty P^s(A_i) : A = \bigcup_{i=1}^\infty A_i\Big\}.$$

Then $\mathcal{P}^s$ is a Borel regular measure on $\mathbf{R}^n$.

That $\mathcal{P}^s$ is a Borel measure can be verified as in the case of Carathéodory's construction in Chapter 4. To see that it is Borel regular, notice first that always $P^s_\delta(\overline{A}) = P^s_\delta(A)$, whence $P^s(\overline{A}) = P^s(A)$. Hence

$$\mathcal{P}^s(A) = \inf\Big\{\sum_{i=1}^\infty P^s(F_i) : A \subset \bigcup_{i=1}^\infty F_i,\ F_i \text{ is closed}\Big\},$$

from which the Borel regularity follows as in 4.2. This last formula also gives for Borel sets $B \subset \mathbf{R}^n$,

$$\mathcal{P}^s(B) = \inf\Big\{\sum_{i=1}^\infty P^s(B_i) : B = \bigcup_{i=1}^\infty B_i,\ B_i\text{s are disjoint Borel sets}\Big\}.$$

Observe also that trivially $\mathcal{P}^s(A) \leq P^s(A)$.

It is evident that

$$\mathcal{P}^t(A) = 0 \quad \text{whenever } \mathcal{P}^s(A) < \infty \text{ and } 0 \leq s < t.$$

Hence the packing measures determine a dimension in the same way as Hausdorff measures. We now show that it is the upper packing dimension defined earlier via the upper Minkowski dimension. This is essentially due to Tricot [2].

5.11. Theorem. *For any* $A \subset \mathbf{R}^n$,

$$\begin{aligned}\overline{\dim}_p A &= \inf\{s : \mathcal{P}^s(A) = 0\} = \inf\{s : \mathcal{P}^s(A) < \infty\} \\ &= \sup\{s : \mathcal{P}^s(A) > 0\} = \sup\{s : \mathcal{P}^s(A) = \infty\}.\end{aligned}$$

Proof. The last four terms are easily seen to equal, and we shall only show

$$\overline{\dim}_p A = d \equiv \inf\{s : \mathcal{P}^s(A) = 0\}.$$

Clearly, for bounded sets $B \subset \mathbf{R}^n$,

$$P(B, \varepsilon/2)\, \varepsilon^s \leq P^s_\varepsilon(B),$$

which leads to $\overline{\dim}_p A \leq d$.

To prove the opposite inequality, let $0 < t < s < d$ and $A_i \subset \mathbf{R}^n$ be bounded with $A = \bigcup_{i=1}^\infty A_i$. It is enough to show that $\overline{\dim}_M A_i \geq t$ for some i. Since $\mathcal{P}^s(A) > 0$, there is i such that $P^s(A_i) > 0$. Let $0 < \alpha < P^s(A_i)$. Then for $\delta > 0$, $P^s_\delta(A_i) > \alpha$ and there exist disjoint closed balls $B_1, B_2, \ldots$ with centres in A_i such that $d(B_j) \leq \delta$ and

$$\sum_j d(B_j)^s \geq \alpha.$$

Assuming $\delta \leq 1$, let for $m = 0, 1, 2, \ldots, k_m$ be the number of the balls B_j for which $2^{-m-1} < d(B_j) \leq 2^{-m}$. Then

$$\sum_{m=0}^\infty k_m 2^{-ms} \geq \sum_j d(B_j)^s \geq \alpha.$$

This yields for some integer $N \geq 0$,

$$2^{Nt}(1 - 2^{t-s})\, \alpha \leq k_N,$$

since otherwise

$$\sum_{m=0}^{\infty} k_m 2^{-ms} < \sum_{m=0}^{\infty} 2^{mt}(1-2^{t-s})\,2^{-ms}\alpha = \alpha.$$

Since $d(B_j) \le \delta$ for all j, we have $2^{-N-1} \le \delta$. Therefore

$$P(A_i, 2^{-N-1}) \ge k_N \ge 2^{Nt}(1-2^{t-s})\,\alpha,$$

which gives

$$\sup_{0<\varepsilon\le\delta} P(A_i,\varepsilon)\,\varepsilon^t \ge P(A_i, 2^{-N-1})\,2^{-Nt-t} \ge 2^{-t}(1-2^{t-s})\,\alpha.$$

Letting $\delta \downarrow 0$, we obtain

$$\limsup_{\varepsilon\downarrow 0} P(A_i,\varepsilon)\,\varepsilon^t > 0,$$

and so $\overline{\dim}_M A_i \ge t$ as required. □

Next we compare packing measures to Hausdorff measures.

5.12. Theorem. *For all $A \subset \mathbf{R}^n$, $\mathcal{H}^s(A) \le \mathcal{P}^s(A)$.*

Proof. It suffices to show $\mathcal{H}^s(A) \le P^s(A)$, and for this we may assume $P^s(A) < \infty$. Let $\varepsilon > 0$ and choose $\delta > 0$ such that $P^s_\delta(A) < P^s(A) + \varepsilon$. Let $B_1, B_2, \ldots$ be disjoint closed balls with centres in A such that $d(B_i) \le \delta$ and

(1) $$\sum_i d(B_i)^s \le P^s_\delta(A) \le \sum_i d(B_i)^s + \varepsilon.$$

Since $P^s_\delta(A) < \infty$, there is k for which

(2) $$\sum_{i=k+1}^{\infty} d(B_i)^s < \varepsilon.$$

We can apply the covering theorem 2.1 to the family of closed balls $B(x,r)$ such that $x \in A$, $10r \le \delta$, and

$$B(x,r) \subset \mathbf{R}^n \setminus \bigcup_{i=1}^{k} B_i.$$

Then we find disjoint closed balls $B_1', B_2', \dots$ of diameter at most $\delta/5$ with centres in A such that

$$(3) \qquad A \setminus \bigcup_{i=1}^{k} B_i \subset \bigcup_j 5B_j'$$

and that the combined collection $\{B_i : i = 1, \dots, k\} \cup \{B_j' : j = 1, 2, \dots\}$ is also disjoint. Hence by (1) and (2)

$$\begin{aligned}\sum_{i=1}^{k} d(B_i)^s + \sum_j d(B_j')^s \leq P_\delta^s(A) &\leq \sum_{i=1}^{\infty} d(B_i)^s + \varepsilon\\ &\leq \sum_{i=1}^{k} d(B_i)^s + 2\varepsilon,\end{aligned}$$

and so

$$\sum_j d(B_j')^s \leq 2\varepsilon.$$

Consequently by (3)

$$\begin{aligned}\mathcal{H}_\delta^s(A) &\leq \sum_{i=1}^{k} d(B_i)^s + \sum_j d(5B_j')^s\\ &= \sum_{i=1}^{k} d(B_i)^s + 5^s \sum_j d(B_j')^s\\ &\leq P_\delta^s(A) + 5^s 2\varepsilon < P^s(A) + (1 + 5^s 2)\,\varepsilon.\end{aligned}$$

Letting $\delta \downarrow 0$ and $\varepsilon \downarrow 0$, we get $\mathcal{H}^s(A) \leq P^s(A)$. □

5.13. Remarks. In Theorem 5.12 the strict inequality is possible even in the sense that $\mathcal{H}^s(A) = 0$ and $\mathcal{P}^s(A) = \infty$. On the other hand, the equality $0 < \mathcal{H}^s(A) = \mathcal{P}^s(A) < \infty$ holds in some sense rather rarely; it forces s to be an integer and A rectifiable in a sense to be defined later. We come to this in Chapter 17. Anyway, for nice integral dimensional sets $\mathcal{H}^s$ and $\mathcal{P}^s$ agree and so $\mathcal{P}^1, \mathcal{P}^2, \dots$ also generalize the classical concepts of length, area,... measures.

Packing measures were introduced by Tricot [2], Taylor and Tricot [1]–[2] and Sullivan [3]. Sullivan showed that the natural measure on limit sets of Kleinian groups sometimes is given as a packing measure; sometimes it is given as a Hausdorff measure. For later results of this type on dynamical systems, see Denker and Urbański [2]. For some other results on packing measures, see Alestalo and Väisälä [1], Cutler [4]–[5], Edgar [2], Haase [1]–[4], Joyce and Preiss [1], Mattila and Mauldin [1], Meinershagen [1], Peres [3], Rezakhanlou [1] and Saint Raymond and Tricot [1].

For other dimensional concepts, some of them related to ergodic and information theory, see e.g. Billingsley [1], Blei [1], Cajar [1], Cutler [2]–[3], Cutler and Olsen [1], Falconer [16], Hu and Lau [2], Nusse and Yorke [1], Olsen [1], Pesin [1]–[2], Reyes and Rogers [1], Rogers [2], Tricot [1] and Walters [1].

Other measures that agree with constant multiples of Hausdorff measures for nice integral dimensional sets are the integralgeometric measures which we shall define here and then briefly discuss their properties without going into the proofs.

Integralgeometric measures

5.14. As a starting point we can take an integralgeometric formula for the length of a rectifiable curve Γ in $\mathbf{R}^2$: for each line L count the number of points in the intersection $\Gamma \cap L$ and integrate this number over all lines $L \subset \mathbf{R}^2$. Doing this with affine $(n-m)$-dimensional subspaces of $\mathbf{R}^n$, we can define for any Borel set $A \subset \mathbf{R}^n$

$$\mathcal{I}_1^m(A) = \iint_V \mathcal{H}^0\big(A \cap P_V^{-1}\{a\}\big)\, d\mathcal{H}^m a\, d\gamma_{n,m} V$$

(recall that $\mathcal{H}^0$ counts the number of points). Then $\mathcal{I}_1^m$ is called the m-dimensional integralgeometric (or Favard) measure (with parameter 1) in $\mathbf{R}^n$. On a smooth m-dimensional surface it agrees with a constant multiple of the Hausdorff measure $\mathcal{H}^m$. For very general integralgeometric formulas involving the measures $\mathcal{I}_1^m$, see Federer [2]. This paper also studies relations to the topological dimension.

It turns out that $\mathcal{I}_1^m$ can also be defined via Carathéodory's construction. At the same time we get a one-parameter family of integralgeometric measures. To do this we choose $\mathcal{F}$ to be the family of all Borel subsets of $\mathbf{R}^n$ and for $1 \le t \le \infty$ we let ζ_t^m be the function defined on Borel sets by

$$\zeta_t^m(B) = \Big(\int \mathcal{H}^m(P_V B)^t d\gamma_{n,m} V\Big)^{1/t}, \quad \text{if } 1 \le t < \infty,$$
$$\zeta_\infty^m(B) = \operatorname{ess\,sup}\{\mathcal{H}^m(P_V B) : V \in G(n,m)\}.$$

(Of course, the essential supremum is taken with respect to $\gamma_{n,m}$.) The $\gamma_{n,m}$ measurability of the function $V \mapsto \mathcal{H}^m(P_V B)$ is not at all clear, but it can be demonstrated with the aid of the theory of Suslin sets, see Federer [3, 2.10.5].

For $t = 1$ the above two definitions of $\mathcal{I}_1^m$ agree, see Federer [3, 2.10.15]. For a fixed m the measures $\mathcal{I}_t^m$, $1 \leq t \leq \infty$, have the same null-sets: $\mathcal{I}_t^m(A) = 0$, if and only if A is contained in a Borel set B such that $\mathcal{H}^m(P_V B) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n,m)$, see Federer [3, 2.10.5]. But in general the relation between these measures for a fixed m and varying t is not yet clear. Mattila [10] constructed a compact subset E of $\mathbf{R}^2$ for which $\mathcal{I}_1^1(A) < \mathcal{I}_t^1(A) = \infty$ for $t > 1$, but it is not known whether $\mathcal{I}_t^m$ is a constant multiple of $\mathcal{I}_\infty^m$ for $1 < t < \infty$.

The question what happens to the null-sets of integralgeometric measures under smooth maps was studied by Mattila [11]. It was shown that a C^2 diffeomorphism $f\colon \mathbf{R}^2 \to \mathbf{R}^2$ preserves the null-sets of $\mathcal{I}_1^1$ if and only if it is affine. The non-trivial part is to construct for a given non-affine C^2 diffeomorphism $f\colon \mathbf{R}^2 \to \mathbf{R}^2$ a compact set $E \subset \mathbf{R}^2$ such that $\mathcal{I}_1^1(E) = 0$ and $\mathcal{I}_1^1(fE) > 0$. The basic idea is the following. Since f is not affine it sends many line segments to curves which are not straight line segments. Consider a parallelogram P as in Figure 5.2. Suppose that the images of the line segments which are parallel to shorter sides of P and join the longer sides are curved. Then we can put many narrow parallel parallelograms P_i inside P in such a way that the length of some projections of $\bigcup_i P_i$ is small but $f(\bigcup_i P_i)$ projects onto the same set as $f(P)$ in every direction. We repeat a similar "Venetian blind" construction inside each P_i turning the direction of the new parallelograms and continue to get the desired Cantor set E. The difficulty is that since f is smooth small line segments are curved very little under f and so the parallelograms must be put very close to each other in order to preserve the projections on the image side. But this has the effect that on the domain side there are very few directions where the projections are small. For overcoming this problem and other details, see Mattila [11].

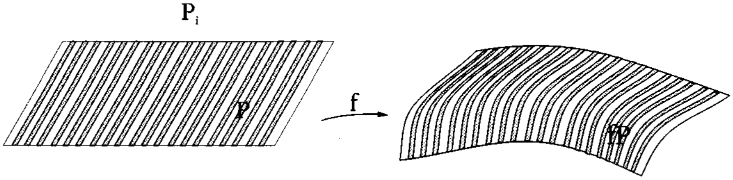

Figure 5.2.

Other closely related integralgeometric concepts are Vitushkin's variations of sets, see Ivanov [1], and those used by Murai [2] in complex analysis.

Exercises.

1. Let $A = \{0\} \cup \{1/i : i = 1, 2, \ldots\}$. Show that $\overline{\dim}_M A = \underline{\dim}_M A = 1/2$.
2. Prove the formulas of 5.3 for $\overline{\dim}_M A$ and $\underline{\dim}_M A$ in terms of $\log N(A, \varepsilon)$.
3. Show that $2^{-s-n}\alpha(n)\,\mathcal{H}^s(A) \leq \mathcal{M}^s_*(A)$.
4. Show that $\overline{\dim}_M(A \cup B) = \max\{\overline{\dim}_M A, \overline{\dim}_M B\}$ for bounded subsets A and B of $\mathbf{R}^n$. Give an example where this fails for $\underline{\dim}_M$.
5. Show that for any $A \subset \mathbf{R}^n$ and $\varepsilon > 0$ there exists an increasing sequence $A_1 \subset A_2 \subset \cdots \subset A$ of bounded sets A_i such that $A = \bigcup_{i=1}^{\infty} A_i$ and $\overline{\dim}_M A_i \leq \dim_p A + \varepsilon$ for all i.
6. Show that for bounded sets $A \subset \mathbf{R}^m$ and $B \subset \mathbf{R}^n$, $\overline{\dim}_M(A \times B) \leq \overline{\dim}_M A + \overline{\dim}_M B$.
7. Construct an example to show that there need not be equality in the preceding exercise.
8. What can be said about the lower Minkowski dimension of cartesian products?